Customer Shopping Behavior Analysis

Python • SQL • Power BI Dashboard

Full End-to-End Data Analysis Project — 3,900 Transactions

Project Overview

Python + Pandas

Data loading, cleaning, feature engineering, export to SQL Server

SQL Server

13 business questions answered using T-SQL queries

Power BI

4-page interactive dashboard with DAX measures and visual storytelling

Dataset Summary

Scale

- 3,900 transactions
- 50 US states
- 25 products
- 18 → 20 engineered features

Customer Profile

- Age range: 18–70 years
- Purchase range: $20–$100
- 68% male, 32% female
- 27% subscribers

Key Finding: Revenue Imbalance

$158K

Male Revenue

68% of total sales

$75K

Female Revenue

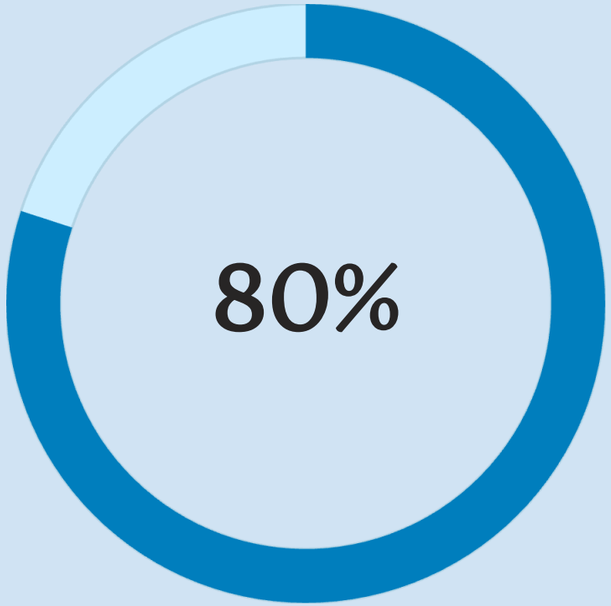
32% of total sales

68%

Gender Split

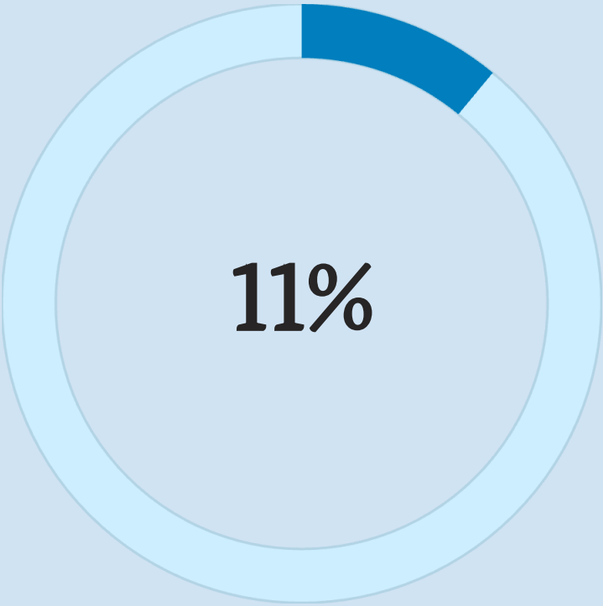
Reflects dataset composition

Customer Segmentation



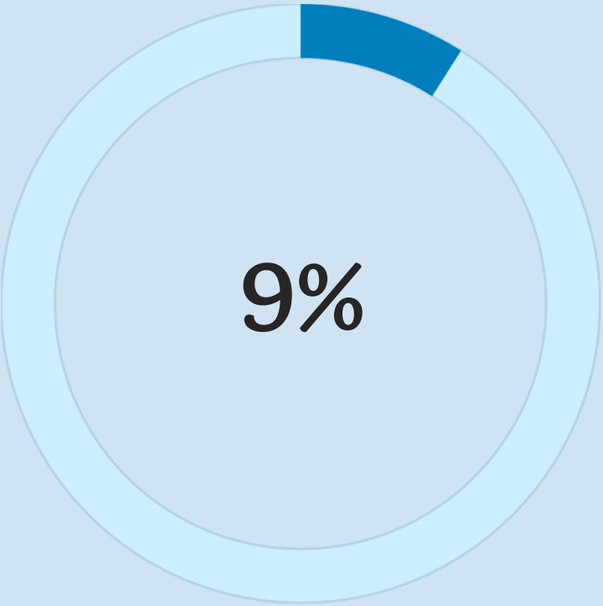
Loyal

3,116 customers with 30+ purchases



New

424 first-time buyers



Returning

360 customers with 6–20 purchases

Retention programs matter more than acquisition

Discount Strategy Insights

Discount Paradox

839 customers (21.5%) used discounts AND spent above average — discounts attract high-value buyers, not just bargain hunters

Top Discount Products

- Hat: 50% discount rate
- Sneakers: 49.66%
- Coat: 49.07%

Age Group Analysis

29%

Middle-Aged

46–60 years: highest revenue $67,711

28%

Adult

31–45 years: $64,927 revenue

24%

Young Adult

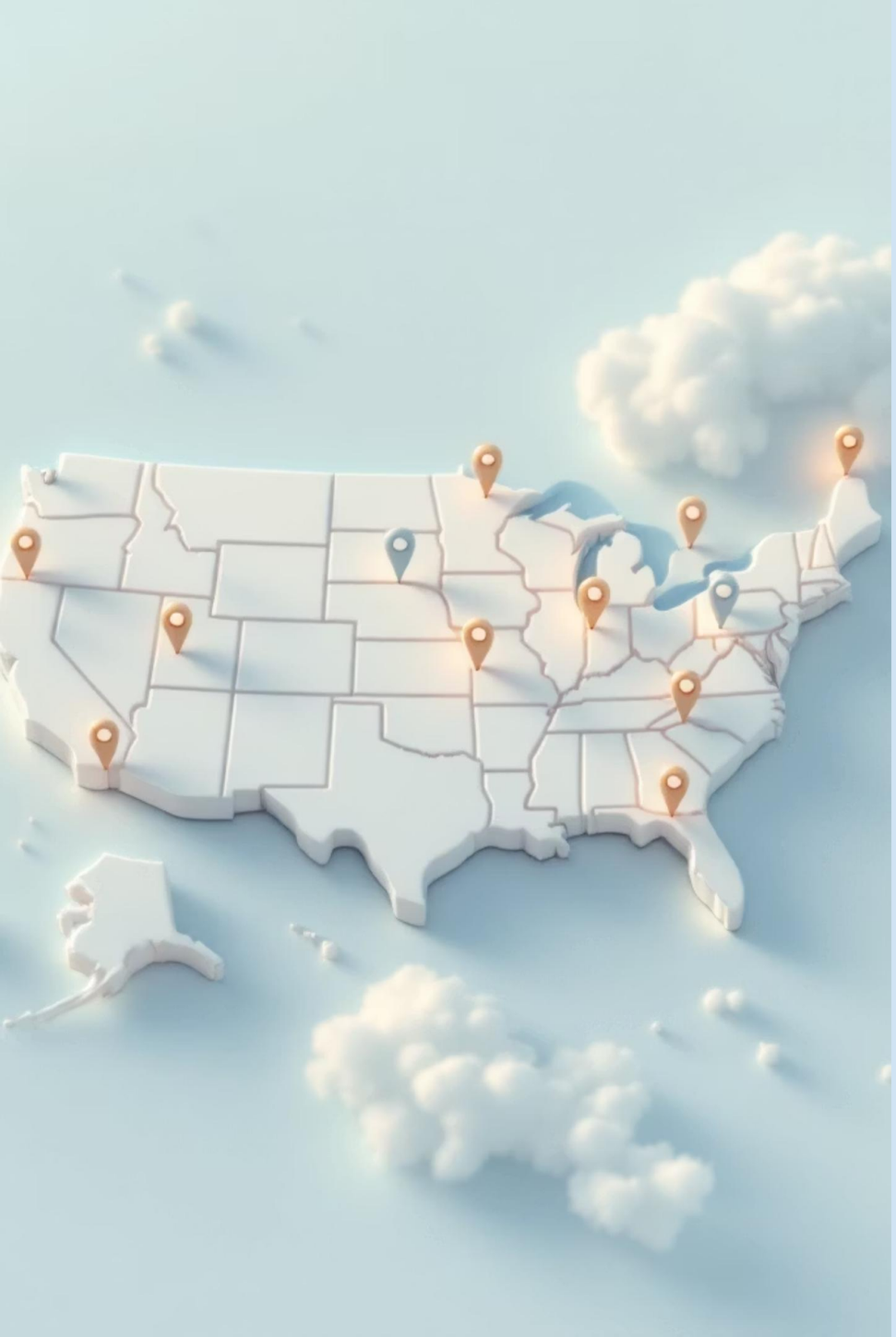
18–30 years: $57,279 revenue

19%

Senior

61–70 years: $43,164 revenue

Young Adults spend most per transaction ($60.36 avg)



Geographic Distribution

$5,784	$3,437	$2.3K
Top State	**Bottom State**	**Revenue Gap**
Montana leads revenue	Kansas lowest revenue	Only $2.3K difference across 50 states

No dominant market — consistent geographic performance

Business Recommendations

01

Boost Subscriptions

Launch loyalty-to-subscription campaign targeting 2,324 Loyal non-subscribers. 10% conversion adds ~232 new subscribers.

02

Smart Discount Strategy

Target 839 high-value discount users with exclusive early-access offers rather than blanket discounts to protect margins.

03

Focus on Middle-Aged

Prioritize 46–60 segment in product placement, email campaigns, and loyalty rewards — generates highest revenue.

04

Promote Top Products

Test raising base price on Hat and Sneakers to reduce discount dependency while maintaining volume.

05

Engage Seniors

Target 61–70 segment with simplified UX and senior-friendly offers — currently underserved with lowest revenue.

Power BI Dashboard

4-Page Interactive Dashboard

Visualizes all 13 SQL findings with unified blue theme and cross-page filters

Key Visualizations

Revenue by gender, age group analysis, product ratings, customer segments, geographic view

Conclusion & Strategic Outlook

This comprehensive, end-to-end data analysis project, utilizing Python, SQL, and Power BI, provides a deep understanding of customer shopping behaviors.

Key Insights

- Significant revenue imbalance by gender
- Strong 80% customer loyalty base
- Discounts attract high-value segments
- Middle-aged customers drive highest revenue
- Geographic distribution is highly consistent

Strategic Focus

- Drive subscription growth
- Implement smart, targeted discounting
- Prioritize high-value customer segments
- Engage underserved demographics (seniors)

These actionable insights will enable optimization of revenue streams, enhance customer retention efforts, and refine overall marketing strategies for future growth.